Supplementary Material for Reviewers: Learned Spectrum Sensors Under Noise-Shape Uncertainty: An Invariant-Detection Analysis

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Abstract

This document is provided for the reviewers of the manuscript "Learned Spectrum Sensors Under Noise-Shape Uncertainty: An Invariant-Detection Analysis." It contains numerical checks of the theory, implementation details, noise synthesis procedures, extended evaluation tables with 95% bootstrap confidence intervals, and additional figures. The manuscript itself, including the proofs in its Appendix, is self-contained.

S-I. Numerical Checks of the Theory

The proofs of all results are in the Appendix of the manuscript. This section reports the numerical checks of the theory.

Theorem 1(ii): uniform achievability for flat-band signals. We simulated the band ratio with a fixed threshold $R + \eta$ (no calibration), η placed halfway between R and the worst-case $\mathcal{H}_1$ limit at one decibel above the wall, for $\rho = 0.169$ and $u = 1$ dB (wall -10.9 dB). The false-alarm rate was taken as the maximum over 42 members of $\mathcal{U}_R$: the two extremal shapes and 40 random two-level and smooth shapes with peak-to-peak ratio exactly R. The detection rate was taken as the minimum over noise shapes c_1 and white. From $N = 512$ to $N = 32\,768$, the maximum false-alarm rate fell from 0.34 to 0.005, and the worst-case detection rate one decibel above the wall rose from 0.63 to 1.00. At the wall itself it fell from 0.34 to 0.00, as expected when the threshold lies above the limiting $\mathcal{H}_1$ ratio at the wall.

Theorem 2: exact wall for arbitrary spectra. (a) A linear program maximises γ over all pairs on a 1024-point grid (variables $(1+\gamma)c_0$, c_1, their minima and γ). It agrees with the closed form of Theorem 2(ii) to within 10^{-7} dB in all 136 cases: 34 spectra (root-raised-cosine signals at 2, 4, 8 and 16 samples per symbol with roll-off 0.15, 0.35 and 0.6, OFDM, flat bands, floored, rippled and random smooth spectra) at $u \in \{0.5, 1, 2, 3\}$ dB. Near the boundary $p_{\max}/p_{\min} = R$ the program is unbounded exactly when the formula predicts an infinite wall (20 of 20 cases). *(b)* Dualising the inner worst-case problems makes the search for the best separating spectral functional a single linear program. Its margin is zero at every tested SNR below the wall and positive at every SNR above it (0.05 to 1 dB on either side), for four spectra. *(c)* For a Gaussian bump on a floor ($u = 1$ dB, exact wall -7.57 dB, while the choice $s = R^2 - 1$ gives -8.29 dB), we used the functional found at one decibel above the wall with a fixed threshold midway in its margin. Over the exact pair, the worst-case shape for the functional and 40 random admissible shapes, the maximum false-alarm rate is 0.40 at $N = 2\,048$ and 0.17 at $N = 32\,768$. The worst-case detection rate one decibel above the wall rises from 0.59 to 0.84, and at the wall itself it falls from 0.32 to 0.01. Convergence is slow because the margin is small, as expected for an asymptotic statement.

Proposition 3: first-order surrogate indistinguishability. With $N = 256$ we drew the ideal surrogate from the estimated covariance of the power-normalised signal windows (4×10^5 windows) and scored signal and surrogate on the same noise draws (10^5 per SNR). For the cyclostationary and fourth-moment detectors the gap is not detectable below -16 dB and then grows like γ^2: gap$/\gamma^2$ stays within 0.30–0.40 for the one-sided fourth moment over -10 to 0 dB. For the LMPI and band-ratio statistics, the gap against the paper's power-normalised surrogates stays within 0.35 points at every SNR. Against the un-normalised ideal surrogate it reaches 1.8 points near $P_d = 0.9$, an $O(\gamma^2)$ effect of the fluctuating window power of a Gaussian process. This is why the surrogates in this paper are power-normalised per window, as the signals are.

Escape law in the wall tests. The derivation in the Appendix of the manuscript neglects the self-noise of the signal, which slightly overestimates P_d near $P_d \approx 0.9$ (largest deviation 0.07). In the wall tests, the robust threshold over $\{\text{white}, c_0, c_1\}$ replaces the nominal χ^2 quantile. The false-alarm rate of CYC hardly depends on the noise shape, so the two differ little; all predictions nevertheless use the measured robust thresholds.

S-II. Implementation Details & Protocols

Noise synthesis. PSD-defined shapes (pink, RTL-like, band tilts, extremal pairs, random smooth shapes) are synthesised in the DFT domain: white Gaussian noise is multiplied by the square root of the shape on the N-point grid and transformed

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back. The resulting process is circulant, and its expected periodogram equals the shape exactly. AR(1) noise is generated by recursion with a stationary start. The Butterworth roll-offs are causal IIR filters applied to $2N$ white samples, of which the last N are kept. This noise is not circulant, which is why per-bin whitening cannot undo the LP .8 edge.

PSD estimates. The signal PSD for the extremal pairs is the Hann-windowed average periodogram of 4000 signal windows, normalised to unit mean. For $N > 2048$ a 2048-point estimate from 8000 windows is interpolated, so that estimation noise does not make the pair inexact. Reference whitening averages the periodograms of K noise windows. It then either fits the log-periodogram with a 16-term cosine and sine series by least squares, excluding DC, or smooths it with a 3-bin circular moving average. Each input's DFT is divided by the square root of the normalised estimate.

Recorded noise. The band-edge roll-off of a recording is the ratio, in dB, of its averaged 128-point Hann periodogram over $|f| < 0.15$ to that over $|f| > 0.42$ (cycles per sample). A recording is flagged as narrowband if its periodogram exceeds a 9-bin median-filtered version by more than 2 dB outside the DC region. All control files have less than 0.5 dB of roll-off. The Receiver 1 dataset states no licence; we use it for evaluation only and do not redistribute it. The Receiver 2 dataset is MIT-licensed. Neither contains recordings made by the author.

Statistics. Confidence intervals for all diagnostic and recipe cells come from a two-level bootstrap ($B = 2000$). Level 1 resamples training seeds with replacement; Level 2 resamples the finite evaluation windows through binomial draws (multinomial for paired comparisons) at the observed per-seed rates. For surrogate gaps, Level 2 uses a normal approximation with standard error $\sqrt{0.5/(3n)}$ for n windows, the largest binomial error of a difference of two detection rates averaged over three SNR points. McNemar tests pool the windows of all seeds and therefore compare the trained models, not the recipes in general; the bootstrap intervals, which also resample seeds, are the recipe-level statement. Paired comparisons between matched and unmatched noise recipes on the same test streams use McNemar's test with Edwards' continuity correction.

Ablations and re-runs. Without power normalisation, the GAP CNN becomes sensitive to noise power again ($P_f = 25\%$ at +1 dB). Changing the training volume by 3× leaves surrogate gaps at −0.4 and +0.0 points and LP .8 false alarms at 33% and 30%. An early-stopped ResNet checkpoint gives 30% false alarms and a gap of +0.5 points. The seed-to-seed variation of surrogate training at $N = 2048$ (0.0–19.8%) was reproduced on a re-run (23.8%).

Escape controls. The noise of the wall tests is synthesised as circulant (periodic) windows, whereas signals are linear segments of longer streams, and the Paired negatives are per-window phase-randomised (also periodic). To exclude that a network separates periodic from non-periodic components, the escape test at $N = 8192$ was repeated with noise windows cut from the middle of four-times-longer realisations (Toeplitz covariance) and with a stationary Gaussian process of the signal's PSD in place of the signal (3 seeds). Detection rates two decibels below the $u = 2$ dB wall are 0.38 (GapCNN-Paired) and 0.61 (CycNet) with linear noise, within 0.03 of the circulant values; the Gaussian process is flagged at most 0.05 of the time at or below the wall.

Capacity and data scaling. Scaling model capacity by 16× (GapCNN-4X, 1.91M parameters) together with 10× more training data (600 000 examples) leaves the surrogate gap near zero (+0.1 points, $P_d(-10\,\mathrm{dB}) = 70\%$), and LP .8 false alarms remain at 31%. Model size and training volume therefore do not explain the spectral shortcut.

DetectNet replica. Our layer-by-layer replica of DetectNet [1], trained on 16QAM as in the released code, detects Gaussian surrogates as often as signals (mean gap +0.8 points; largest difference over $[-20, 0]$ dB 1.3–2.6 points across seeds) and false-alarms in 24% of LP .8 noise windows. We did not reproduce the published detection curve itself.

OFDM and multipath fading. On OFDM ($\rho = 0.75$, 64 subcarriers, 16-sample CP) through 4-tap Rayleigh fading with CFO (5 seeds), the GAP CNN's surrogate gap is +0.9 points and its P_d at −10 dB is 13%, against 30% for the energy detector with known noise power, consistent with Corollary 1 of the main paper; LP .8 and RTL roll-offs give $P_f = 100\%$ and 53%.

LMPI replay. The evaluation draws every test window from a seeded stream in a fixed order, and score functions consume no randomness. We therefore recomputed the LMPI statistic on exactly the windows of each saved detector by re-running the evaluation with the run's own settings (observation length, samples per symbol, seed, modulations and evaluation sizes), and checked that the replayed scores reproduce themselves key by key. The template is the signal's mean periodogram on the N-point grid; for $N = 8192$ the 2048-point estimate is interpolated. Spectral templates of other detectors are ridge-regression coefficients ($\lambda = 10^{-6}\,\mathrm{tr}(\mathbf{X}^T\mathbf{X})/128$) of their scores on the normalised periodogram, aggregated to 128 bands, over pooled noise, signal and surrogate windows. Since the rows of $\mathbf{X}$ sum to one, only the centred template is identifiable. Real signals were replayed in the same way: all 251 captures reproduce the published band-ratio counts exactly. Fitting the templates of the GAP CNN, CNN–LSTM and covariance CNN within the MPI family of Proposition 4 (bin weights $\gamma_0 p_k/(1+\gamma_0 p_k)$, γ_0 from −40 to +20 dB) gives best design SNRs between the LMPI limit (1 models) and −10 dB. The cosine gain over the LMPI limit is small, so the design SNR is only weakly identified.

Off-band sweep. Signals with carrier offsets of 0 to 0.35 cycles/sample and 4, 8 or 16 samples per symbol were scored at −10, −6 and −2 dB in white noise ($N = 128$, 2000 windows per cell), with nominal thresholds. Each detector's prediction is its own centred-signal curve evaluated at the overlap-scaled SNR.

Training dynamics. GAP CNN, $N = 128$, White, $\mathrm{Surr}^*_\mathrm{m}$ and Paired, seeds 0–2, with the training data and optimiser of the main runs. A fixed probe set (4000 windows each of noise, signals and surrogates at −10 and 0 dB) is scored at log-spaced steps.

TABLE S-I

SENSITIVITY TO THE FALSE-ALARM TARGET α ($N = 128$, WHITE-TRAINED NETWORKS, RANGES OVER THE FIVE ARCHITECTURES; SEED MEANS). THRESHOLDS FROM 2000 HELD-OUT WHITE-NOISE TEST DRAWS; P_f IN %. GAP: LARGEST $|P_d^{\text{signal}} - P_d^{\text{surrogate}}|$ IN POINTS.

	White-trained networks							Band ratio	
α (%)	P_f white	+1 dB	LP .8	RTL	Pink	Gap	$P_d(-10)$	LP .8	$P_d(-10)$
1	0.9–1.1	0.9–1.2	8–16	4–5	100	1.7	27–47	15	42
5	4.6–5.1	4.8–5.2	21–39	11–16	100	0.9	48–69	33	66
10	9.4–9.9	9.7–10.3	32–53	19–26	100	0.4	60–80	46	77

TABLE S-II

TRAINING RECIPES ($N = 128$). LEFT: P_f (%) AT THE NOMINAL THRESHOLD ON COLOURED NOISE. RIGHT: P_d (%) AT -10 DB WITH THE SIGNAL IN WHITE (W), PINK OR LP .8 NOISE AT THE NOMINAL THRESHOLD, AND WORST CASE OVER THE BROAD SET WITH $P_f \leq 5\%$ ENFORCED (WC). SUBSCRIPT M: POSITIVES DRAWN IN THE SAME NOISE MIXTURE AS THE NEGATIVES. MEAN OVER 10/5 SEEDS.

		P_f on noise			P_d, signal in				
Net	Recipe	Pink	LP .8	RTL	W	Pink	LP .8	wc	Gap
CNN–LSTM	WHITE	100	34	16	69	100	90	0	−0.1
	PINK	1	8	9	58	2	39	1	+0.4
	PINK_m	29	1	2	27	43	5	0	+0.8
GAP CNN	WHITE	100	34	16	69	100	90	0	+0.1
	PINK	0	12	15	68	0	30	0	+0.3
	PINK_m	7	6	12	65	25	23	2	+0.9
	RAND	87	0	3	54	93	1	0	+1.1
	RAND_m	100	1	3	57	100	17	0	+1.0
	SURR	5	2	7	54	9	7	4	+4.6
	SURR^*	8	4	9	55	8	14	4	+11.6
	SURR^*_m	19	6	6	54	19	41	3	+9.6

Power versus shape. Power uncertainty is $|10\log_{10}(P_B/P_A)|$ for the mean powers of the two blocks or captures; shape uncertainty is the peak-to-peak ratio of the two smooth shape estimates of the empirical-uncertainty analysis. The energy-detector wall uses (1) of the main paper with this peak-to-peak value. Receiver gain is not recorded in the datasets and is assumed fixed.

MPI efficiency. The MPI statistic of Proposition 4 at each SNR uses the covariance of 4×10^5 power-normalised signal windows ($N = 128$), with thresholds set on 2×10^5 white windows and 10^5 test windows per SNR. The SNR loss of a network is the difference between the SNR at which it reaches a given P_d and the SNR at which the MPI test, scored on the same kind of normalised signal windows, reaches it. The strict Gaussian ceiling (MPI on un-normalised Gaussian surrogates) is not an upper bound for power-normalised windows: at -10 dB it is 0.686, against 0.709 for the MPI test on normalised signals. This is the window-power effect seen in the check of Proposition 3. Under noise-shape uncertainty the ceiling of the worst-case detection rate is α at and below the wall (Proposition 1). Every detector whose worst-case rate exceeds it two decibels below the wall in our escape tests (14 detector–configuration pairs) therefore uses non-Gaussian structure.

S-III. SUPPLEMENTARY TABLES

TABLE S-III

REAL OVER-THE-AIR SIGNALS MIXED INTO REAL NOISE OF THE SAME RECEIVER ($N = 128$; THRESHOLDS AT $P_f = 5\%$ ON THAT RECEIVER'S NOISE). CAPTURES: USED/DOWNLOADED (A CAPTURE IS USED IF IT HAS AT LEAST 20 WINDOWS 6 DB ABOVE ITS NOISE FLOOR IN ITS OCCUPIED BINS). OWN SNR: MEDIAN IN-BAND SNR OF THE SELECTED SIGNAL WINDOWS (DB), I.E. THE SHARE OF THE CAPTURE'S OWN RECEIVER NOISE. MAX P_d: LARGEST DETECTION RATE OVER $\gamma \in [-24, 4]$ DB FOR THE WHITE-TRAINED GAP CNN AND THE CENTRED BAND RATIO. GAP: $P_d^{\text{signal}} - P_d^{\text{surrogate}}$ IN POINTS, MEAN OVER SNRS WITH $0.1 < P_d < 0.9$ (CONTEXT FOURIER SURROGATES); N.D.: NOT DETECTED (MAX $P_d \le 0.1$), GAP UNDEFINED. STRUCTURE STATISTIC: NORMALISED FOURTH MOMENT, ONE-SIDED (κ_{lo}, CONSTANT ENVELOPE) OR TWO-SIDED ($|\kappa - 2|$, PULSED); NEGATIVE VALUES ARISE WHEN THE CAPTURE'S OWN HEAVY-TAILED RECEIVER NOISE DOMINATES (LOW OWN SNR). NETWORKS: MEAN OVER 3 SEEDS.

Class	Receiver	Mod.	Captures	Windows	Own SNR	Max P_d CNN / band	Gap CNN	Gap band	Gap structure	Gap SURR*
FM broadcast	R820T2	WBFM	40/50	4,000	6	0.99 / 0.99	+1.3	+0.7	+23.8 (κ_{lo})	+16.7
FM broadcast	R828D	WBFM	40/40	4,000	11	0.13 / 0.12	0.0	0.0	+20.9 (κ_{lo})	n.d.
NOAA weather	R828D	NBFM	40/40	4,000	1	0.05 / 0.05	n.d.	n.d.	−8.1 (κ_{lo})	n.d.
FRS/GMRS	R828D	NBFM	40/40	4,000	−6	0.07 / 0.06	n.d.	n.d.	−4.0 (κ_{lo})	n.d.
Pager	R828D	FSK	40/40	4,000	−3	0.05 / 0.05	n.d.	n.d.	−17.6 (κ_{lo})	n.d.
ISM sensors	R820T2	FSK/OOK	22/50	2,200	1	0.85 / 0.83	+2.9	+3.1	+22.3 (κ_{lo})	0.0
ISM sensors	R828D	FSK/OOK	5/40	500	9	0.83 / 0.83	+1.1	+1.6	−2.8 (κ_{lo})	+0.8
ADS-B	R820T2	PPM	24/50	2,371	2	0.32 / 0.32	+5.0	+5.6	+14.2 ($\lvert\kappa - 2\rvert$)	n.d.
APRS	R828D	AFSK	0/40	0	no capture with a detectable signal					

TABLE S-IV

FALSE ALARMS (%) AFTER WHITENING WITH K REFERENCE NOISE WINDOWS OF THE SAME SHAPE ($N = 128$; MEAN (MAX) OVER 6 WHITE-TRAINED NETWORKS). NOMINAL $P_f = 5\%$.

PSD estimate	K	White	LP .8	RTL	AR+.2	Pink	Tilt
cosine fit	16	7.1 (10)	12.2 (24)	8.4 (11)	8.7 (11)	7.0 (13)	3.4 (5)
cosine fit	64	4.5 (6)	12.4 (16)	6.1 (9)	4.2 (5)	7.4 (11)	5.7 (6)
cosine fit	256	5.5 (6)	10.9 (15)	5.4 (6)	5.0 (6)	5.9 (7)	4.4 (5)
cosine fit	1024	5.7 (7)	11.7 (14)	5.1 (6)	5.4 (6)	6.4 (7)	4.3 (5)
3-bin average	16	4.6 (6)	20.9 (36)	4.4 (5)	7.4 (12)	2.6 (3)	5.9 (9)
3-bin average	1024	5.1 (5)	13.9 (18)	5.3 (6)	5.7 (6)	4.3 (5)	4.9 (5)

TABLE S-V

COMPUTATIONAL COMPLEXITY AND SPECIFICATIONS OF LEARNED DETECTORS ($N = 128$). ALL INPUTS ARE POWER-NORMALISED PER EXAMPLE. LATENCY: FORWARD PASS OF THE NETWORK FOR ONE WINDOW (BATCH 1), IN MS, ON AN INTEL CORE I5-13420H CPU / RTX 3050 GPU, EXCLUDING INPUT PREPROCESSING (FOR CYCNET, THE CYCLIC FEATURES). COVARIANCE: 16×16 SMOOTHED SAMPLE COVARIANCE. STFT: LOG-POWER, 16-POINT HANN WINDOW, HOP 8. TRAINING: ADAM, BINARY CROSS-ENTROPY, COSINE LEARNING-RATE DECAY, 60 K EXAMPLES (HALF SIGNALS), 12 EPOCHS ($N = 128$), BATCH 128, LEARNING RATE 10^{-3} (3×10^{-4} FOR THE CNN–LSTM).

Detector	Params	MFLOPs	Latency (ms) (CPU / GPU)	Input	Ref.
GAP CNN	120 k	19.2	1.11 / 0.63	I/Q, any N	ours
GAP CNN 4X	1907 k	303.4	3.62 / 0.56	I/Q	ours (scaled)
DetectNet replica	1301 k	11.5	2.49 / 0.80	I/Q	code of [1]
CNN–LSTM	151 k	9.6	5.71 / 0.98	I/Q	[1]
1-D ResNet	264 k	15.1	2.34 / 1.80	I/Q	[2]
Covariance CNN	33 k	7.4	0.49 / 0.39	covariance	[3]
STFT CNN	32 k	6.6	0.47 / 0.46	STFT	[4]
Transformer	136 k	4.2	1.52 / 1.33	I/Q (16 tokens)	ours (breadth)
CycNet	12 k	0.3	0.32 / 0.41	cyclic lags	ours (escape)

TABLE S-VI

CLASSICAL ADAPTIVE BASELINES ($N = 128$, $K = 1024$ REFERENCE WINDOWS) ON RECORDED NOISE FROM TWO DISTINCT RECEIVERS. NOMINAL $P_f = 5\%$. MEAN P_f AND MEAN $P_d\,(-10\,\text{dB})$ OVER CLEAN RECORDINGS. RECEIVER 1: NOOELEC NESDR (R820T2 TUNER); RECEIVER 2: RTL-SDR BLOG V4 (R828D TUNER).

		Receiver 1 (R820T2)		Receiver 2 (R828D)	
Detector	Invariance Property	P_f (%)	P_d (%)	P_f (%)	P_d (%)
AMF [5]	CFAR (hom. noise)	12.5	52.8	6.6	53.1
ACE [6]	Scale-inv. CFAR	5.5	50.7	4.9	49.7
Kelly [7]	GLRT CFAR	11.7	53.0	6.2	53.4
ED-ref	Power-ref. ED	18.4	33.8	13.5	32.0

TABLE S-VII

MAIN EVALUATION WITH 95% TWO-LEVEL BOOTSTRAP CONFIDENCE INTERVALS OVER SEEDS AND TEST WINDOWS ($N = 128$, WHITE-NOISE TRAINING). FORMAT: MEAN [2.5%, 97.5%] ($B = 2000$). GAP: PERCENTAGE POINTS; ALL OTHER COLUMNS: %.

Detector	n	Gap	+1 dB	LP.9	LP.8	RTL	AR+	AR−	Pink	P_d	P_d^{wc}
CNN–LSTM	5	−0.1 [−0.7, +0.6]	5.4 [5.1, 5.8]	14.6 [13.4, 15.9]	33.9 [30.6, 37.4]	16.0 [15.2, 16.9]	26.6 [25.8, 27.4]	0.5 [0.3, 0.6]	100.0 [100.0, 100.0]	69.0 [67.6, 70.3]	23.6 [19.9, 27.0]
DetectNet replica	5	+0.8 [+0.1, +1.5]	5.0 [4.6, 5.4]	11.4 [10.7, 12.0]	23.8 [22.0, 25.5]	12.2 [11.7, 12.7]	19.0 [18.3, 19.6]	1.2 [0.9, 1.4]	100.0 [100.0, 100.0]	54.0 [52.5, 55.6]	22.9 [20.4, 25.7]
1-D ResNet	5	+0.5 [−0.2, +1.1]	5.2 [4.7, 5.6]	10.7 [9.6, 12.2]	21.3 [19.0, 23.5]	11.2 [10.7, 11.7]	17.4 [16.8, 17.9]	1.1 [0.8, 1.3]	100.0 [100.0, 100.0]	48.3 [47.3, 49.5]	18.3 [16.8, 19.9]
Covariance CNN	5	−0.1 [−0.8, +0.7]	5.1 [4.7, 5.6]	12.1 [11.3, 12.9]	25.2 [23.2, 27.0]	15.2 [14.3, 16.3]	25.1 [24.5, 25.8]	0.5 [0.4, 0.6]	100.0 [100.0, 100.0]	67.4 [66.4, 68.5]	29.3 [26.3, 32.3]
STFT CNN	5	+0.8 [+0.2, +1.4]	5.0 [4.6, 5.4]	14.9 [13.2, 17.4]	39.8 [28.1, 48.6]	16.0 [15.2, 17.0]	25.1 [24.2, 26.1]	0.5 [0.4, 0.6]	100.0 [100.0, 100.0]	57.0 [55.7, 58.3]	17.4 [12.9, 25.0]
GAP CNN (ours)	5	+0.1 [−0.6, +0.7]	5.2 [4.9, 5.6]	14.5 [13.4, 15.6]	34.0 [30.8, 36.8]	15.9 [15.2, 16.8]	26.2 [25.5, 26.9]	0.4 [0.3, 0.6]	100.0 [100.0, 100.0]	69.4 [68.5, 70.3]	23.2 [20.6, 26.0]
ED (known σ^2)	3	+0.5 [−0.5, +1.5]	83.2 [82.4, 83.9]	5.9 [5.3, 6.6]	6.7 [6.0, 7.4]	5.2 [4.6, 5.8]	5.2 [4.8, 5.6]	5.0 [4.5, 5.6]	20.4 [19.6, 21.1]	30.0 [28.7, 31.4]	22.6 [21.3, 23.8]
MME	3	+0.1 [−0.7, +0.9]	4.9 [4.5, 5.3]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	17.9 [16.5, 19.1]	11.9 [11.2, 12.6]	12.3 [11.7, 12.9]	100.0 [100.0, 100.0]	20.4 [19.6, 21.3]	0.0 [0.0, 0.0]
Spectral flatness	3	+0.3 [−0.5, +1.0]	5.1 [4.6, 5.5]	91.6 [91.0, 92.1]	100.0 [100.0, 100.0]	14.8 [13.7, 15.8]	11.5 [10.8, 12.1]	11.1 [10.5, 11.7]	100.0 [100.0, 100.0]	15.8 [14.9, 16.7]	0.0 [0.0, 0.0]
Band ratio (oracle)	3	−0.3 [−1.1, +0.4]	5.2 [4.8, 5.6]	14.7 [14.0, 15.3]	33.5 [32.6, 34.3]	17.5 [16.7, 18.2]	27.0 [26.1, 27.9]	0.4 [0.2, 0.5]	100.0 [100.0, 100.0]	66.0 [65.1, 66.9]	18.5 [17.2, 19.8]
Cyclostationary	3	+9.3 [+8.5, +10.0]	4.8 [4.2, 5.3]	5.5 [5.0, 6.0]	8.7 [7.9, 9.5]	5.8 [5.3, 6.3]	5.7 [5.3, 6.2]	6.3 [5.7, 6.9]	15.0 [14.2, 15.8]	5.7 [5.1, 6.2]	2.6 [2.2, 3.1]

TABLE S-VIII

TRAINING RECIPES WITH 95% TWO-LEVEL BOOTSTRAP CONFIDENCE INTERVALS ($N = 128$, $B = 2000$). FORMAT: MEAN [2.5%, 97.5%]. LEFT COLUMNS: P_f ON NOISE SHAPES (%). RIGHT COLUMNS: P_d AT −10 DB (%) AND SURROGATE GAP (PP).

Net	Recipe	n	Pink	LP.8	RTL	W	Pink	LP.8	wc	Gap
CNN–LSTM	WHITE	5	100.0 [100.0, 100.0]	33.9 [30.6, 37.2]	16.0 [15.2, 17.0]	69.0 [67.6, 70.3]	100.0 [100.0, 100.0]	89.8 [88.1, 91.7]	0.0 [0.0, 0.0]	−0.1 [−0.7, +0.6]
	PINK	5	0.9 [0.4, 1.5]	7.8 [3.3, 12.3]	8.7 [5.4, 11.8]	57.9 [52.2, 64.0]	2.0 [1.1, 3.1]	39.3 [23.0, 55.9]	0.8 [0.3, 1.4]	+0.4 [−0.2, +1.0]
	$\mathrm{PINK_m}$	5	45.2 [14.5, 78.6]	2.3 [0.0, 5.5]	3.4 [1.3, 5.5]	27.9 [19.4, 34.7]	58.3 [31.7, 85.0]	9.0 [1.2, 17.4]	0.1 [0.0, 0.2]	+0.8 [−0.0, +1.7]
GAP CNN	WHITE	5	100.0 [100.0, 100.0]	34.0 [30.9, 36.8]	15.9 [15.2, 16.8]	69.4 [68.6, 70.2]	100.0 [100.0, 100.0]	90.4 [89.0, 91.9]	0.0 [0.0, 0.0]	+0.1 [−0.6, +0.7]
	PINK	5	0.2 [0.1, 0.3]	12.2 [7.4, 17.7]	15.5 [14.5, 16.6]	68.3 [67.2, 69.4]	0.4 [0.2, 0.6]	30.3 [20.1, 41.2]	0.1 [0.0, 0.1]	+0.3 [−0.4, +1.0]
	$\mathrm{PINK_m}$	5	7.0 [6.4, 7.6]	5.6 [3.1, 8.9]	11.7 [10.1, 13.1]	65.4 [63.6, 67.0]	24.6 [22.5, 26.9]	23.1 [15.2, 32.3]	1.8 [1.2, 2.4]	+0.9 [+0.2, +1.6]
	RAND	5	87.0 [84.3, 90.2]	0.0 [0.0, 0.1]	2.6 [2.3, 2.9]	54.3 [53.2, 55.4]	92.7 [90.6, 95.0]	0.5 [0.3, 0.7]	0.0 [0.0, 0.0]	+1.1 [+0.3, +1.8]
	$\mathrm{RAND_m}$	5	100.0 [100.0, 100.0]	0.5 [0.4, 0.7]	2.7 [2.4, 3.0]	57.0 [56.0, 57.9]	100.0 [100.0, 100.0]	17.2 [14.2, 20.9]	0.0 [0.0, 0.0]	+1.0 [+0.4, +1.6]
	SURR	5	5.4 [4.4, 6.4]	1.6 [1.0, 2.1]	7.2 [6.7, 7.8]	54.1 [53.2, 55.0]	8.7 [6.6, 10.6]	7.2 [6.1, 8.3]	4.5 [3.6, 5.2]	+4.6 [+4.0, +5.3]
	SURR^*	5	7.9 [7.0, 8.6]	4.2 [3.1, 5.4]	8.5 [8.1, 9.0]	55.0 [53.1, 56.7]	8.3 [7.3, 9.5]	14.5 [11.1, 17.7]	4.4 [3.7, 5.1]	+11.6 [+7.7, +15.5]
	$\mathrm{SURR}^*_{\mathrm{m}}$	5	18.8 [16.9, 20.4]	5.6 [4.9, 6.2]	6.1 [5.4, 6.8]	53.6 [51.4, 55.5]	18.9 [17.7, 20.1]	41.3 [37.7, 45.0]	3.0 [2.3, 3.6]	+9.6 [+6.6, +12.8]

S-IV. CONFIDENCE INTERVALS AND PAIRED TESTS

Two-level bootstrap intervals for the main and recipe tables and the paired comparisons quoted in Section VII of the main paper (method in Section S-II).

Paired tests on identical test windows confirm the gains of matched colouring for the trained models (McNemar tests pool the windows of all seeds; the bootstrap intervals also resample seeds). On LP.8 at −10 dB, $\mathrm{SURR}^*_{\mathrm{m}}$ gains +26.8 percentage points over SURR^* (95% bootstrap CI [+20.2, +33.5] points, McNemar $\chi^2 = 2333.8$, $p < 10^{-300}$), and $\mathrm{RAND_m}$ gains +16.7 points over RAND ([+13.9, +20.3] points, $\chi^2 = 1667.0$, $p < 10^{-300}$). On pink noise, $\mathrm{PINK_m}$ gains +24.2 points for GAP CNN ([+22.1, +26.4] points, $\chi^2 = 2420.0$, $p < 10^{-300}$) and +56.3 points for CNN–LSTM ([+30.7, +82.6] points, $\chi^2 = 5620.0$, $p < 10^{-300}$).

TABLE S-IX

PAIRED HYPOTHESIS TESTS AND TWO-LEVEL BOOTSTRAP CONFIDENCE INTERVALS FOR MATCHED VERSUS UNMATCHED TRAINING RECIPES ($N = 128$, -10 dB NOMINAL THRESHOLD, $B = 2000$). GAIN: P_d INCREASE IN PERCENTAGE POINTS [95% CI]. b/c: DISCORDANT TEST WINDOWS. MCNEMAR χ^2 WITH CONTINUITY CORRECTION.

Comparison	Noise condition	Windows	Gain (pp) [95% CI]	b/c	χ^2	p-value
GapCNN SURR$^*_m$ vs SURR*	lpf0.8	10,000	+26.8 [+20.2, +33.5]	2876 / 197	2333.8	$< 10^{-300}$
GapCNN RAND$_m$ vs RAND	lpf0.8	10,000	+16.7 [+13.9, +20.3]	1669 / 0	1667.0	$< 10^{-300}$
GapCNN PINK$_m$ vs PINK	pink	10,000	+24.2 [+22.1, +26.4]	2422 / 0	2420.0	$< 10^{-300}$
CLDNN PINK$_m$ vs PINK	pink	10,000	+56.3 [+30.7, +82.6]	5631 / 3	5620.0	$< 10^{-300}$

S-V. SUPPLEMENTARY FIGURES

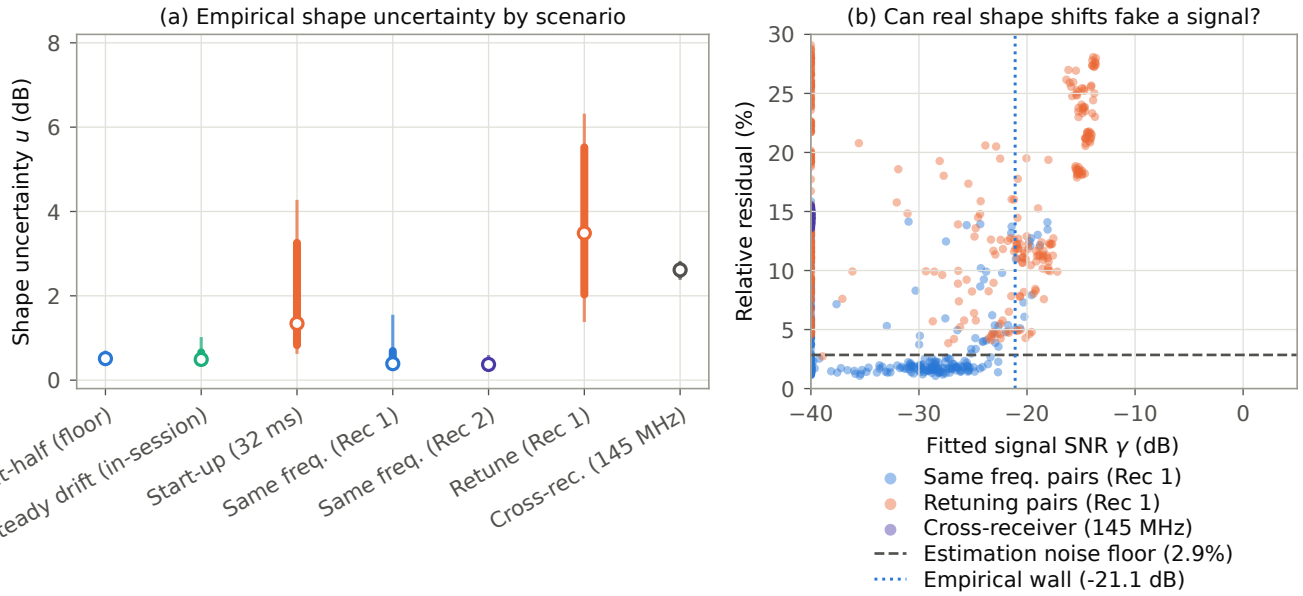


Fig. S-1. Empirical noise-shape uncertainty of two physical SDR receivers (Receiver 1: R820T2; Receiver 2: R828D) over several tuning sessions and frequencies.

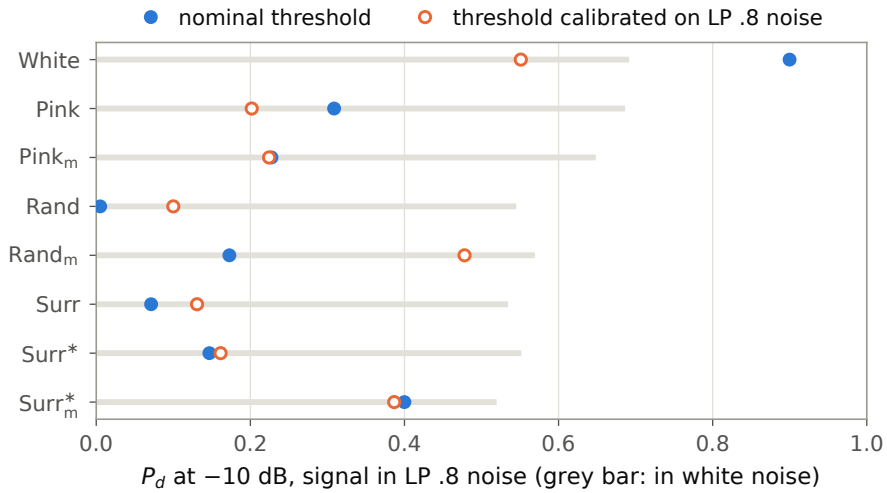


Fig. S-2. Masking of the primary signal under the coloured-noise training recipes.

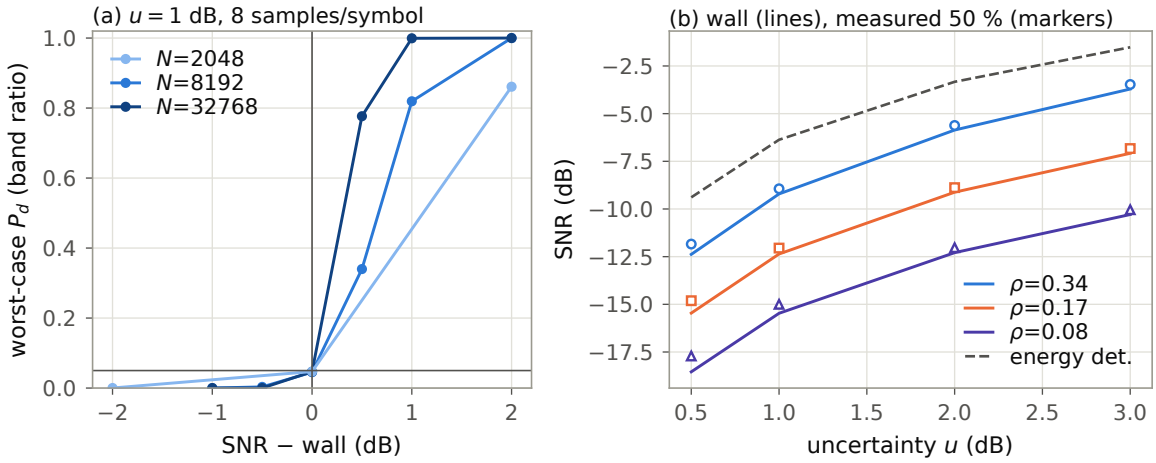


Fig. S-3. Theoretical noise-shape SNR wall against the empirical 50% detection crossing over 24 configurations.

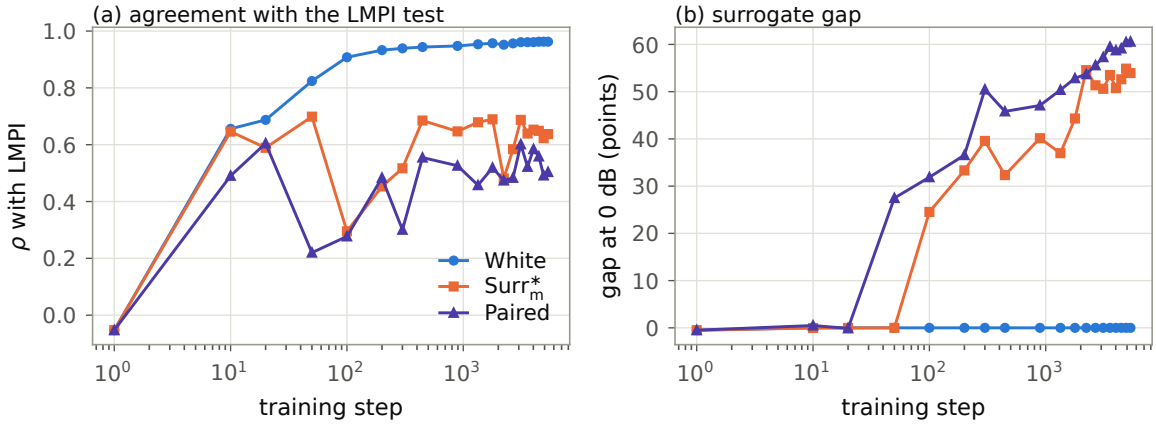


Fig. S-4. Training dynamics of the GAP CNN (N = 128, mean of 3 seeds) under the WHITE, SURR$^*_m$ and PAIRED recipes: (a) Spearman correlation with the LMPI statistic on signal windows at −10 dB; (b) surrogate gap at 0 dB. Every recipe first learns the second-order statistic; only the recipes that remove the spectral cue later acquire structure, and mainly at high SNR.

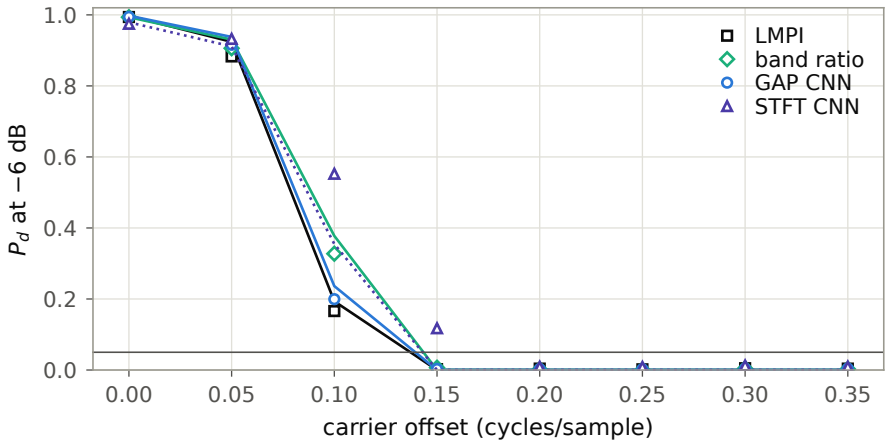


Fig. S-5. Detection of shifted signals (8 samples per symbol, −6 dB) against carrier offset. Markers: measured (GAP CNN: mean of 3 seeds). Lines: predicted from each detector's template overlap, without fitting (dotted: STFT CNN, whose log-spectral input is not captured by a linear template). Negative overlap pushes the detection rate below the false-alarm level (grey line).

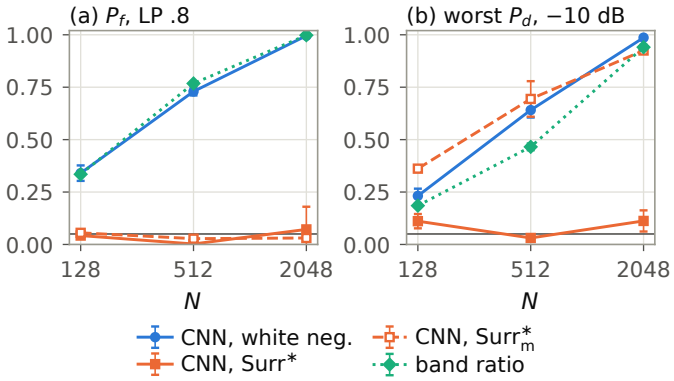


Fig. S-6. Observation-length scaling (Sections VII-C and VII-I of the manuscript). (a) False alarms at the nominal threshold under the 0.8-Nyquist roll-off. (b) Worst-case P_d at -10 dB over the receiver uncertainty set with $P_f \leq 5\%$ enforced. Mean ± s.d. over seeds.

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